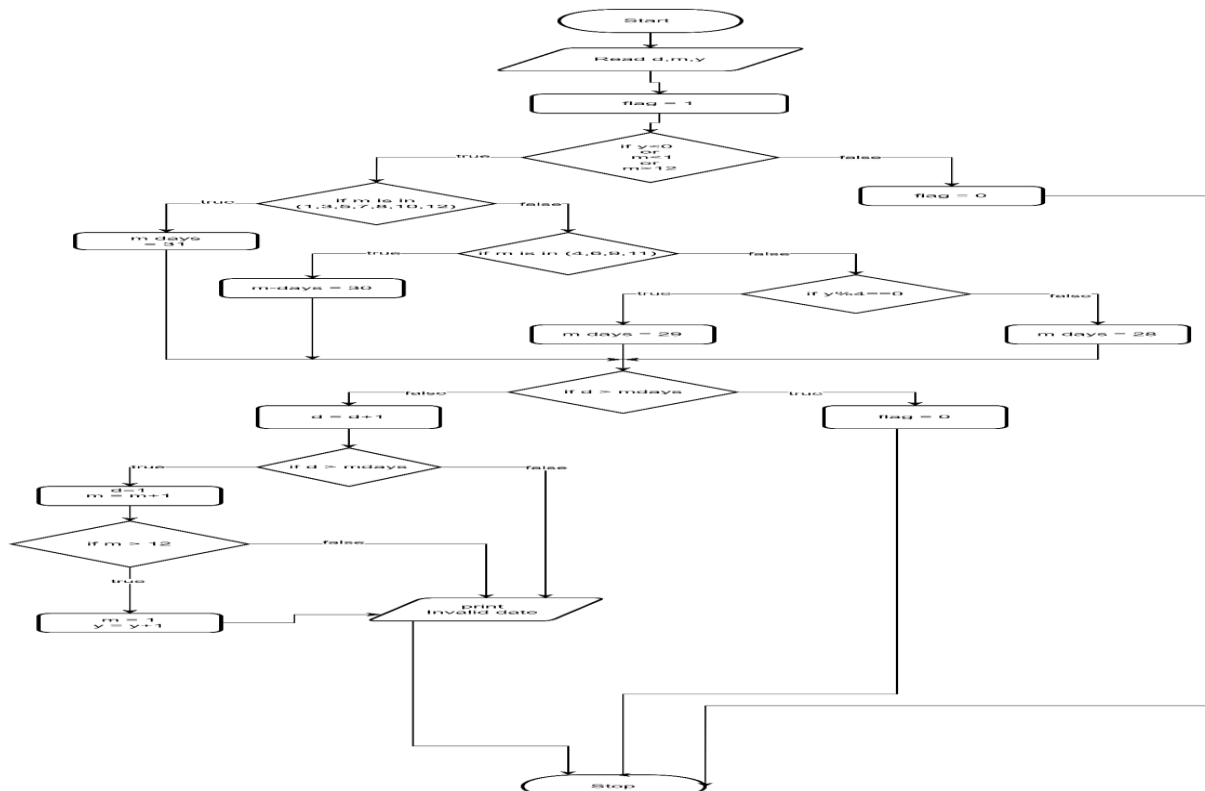


EX 6.1.1-INCREMENTED DATE

ALGORITHM:

1. Input: day, month, and year.
2. Check if year > 0.
 - Check if month is between 1 and 12.
 - Determine max days for the month (considering Leap Year for February).
 - Check if day is between 1 and max days.
 - If any check fails, print "Invalid Date" and stop.
3. Increment Logic:
 - New Year(If day =31 and month = 12) ◦ Set day = 1, month = 1, year = year + 1.
 - New Month(Else if day = max days:) ◦ Set day = 1, month = month + 1.
 - Next Day(Else:) ◦ Set day = day + 1.
4. Output: Print the updated day, month, and year.

FLOWCHART:



6.1.1. Incremented Date

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

1. Valid Month: 1 to 12

2. Valid Day: Depends on the month and year

- January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
- April (4), June (6), September (9), November (11): 1 to 30 days
- February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year

3. Valid Year: Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Output Format:

- If the date is valid, print the incremented date in the format:

```
<DD>-<MM>-<YYYY>
```

- If the date is invalid, print: "Invalid Date".

nextDate.py

```
1 day = int(input())
2 month = int(input())
3 year = int(input())
4 def is_leap(y):
5     if (y % 400 == 0) or (y % 4 == 0 and y % 100 != 0):
6         return True
7     return False
8     if year <= 0:
9         print("Invalid Date")
10    elif month < 1 or month > 12:
11        print("Invalid Date")
12    else:
13        if month in [1, 3, 5, 7, 8, 10, 12]:
14            max_days = 31
15        elif month in [4, 6, 9, 11]:
16            max_days = 30
17        elif month == 2:
18            if is_leap(year):
19                max_days = 29
20            else:
21                max_days = 28
```

```
15
03
2026
16-03-2026
```

```
=== YOUR PROGRAM HAS ENDED ===
```